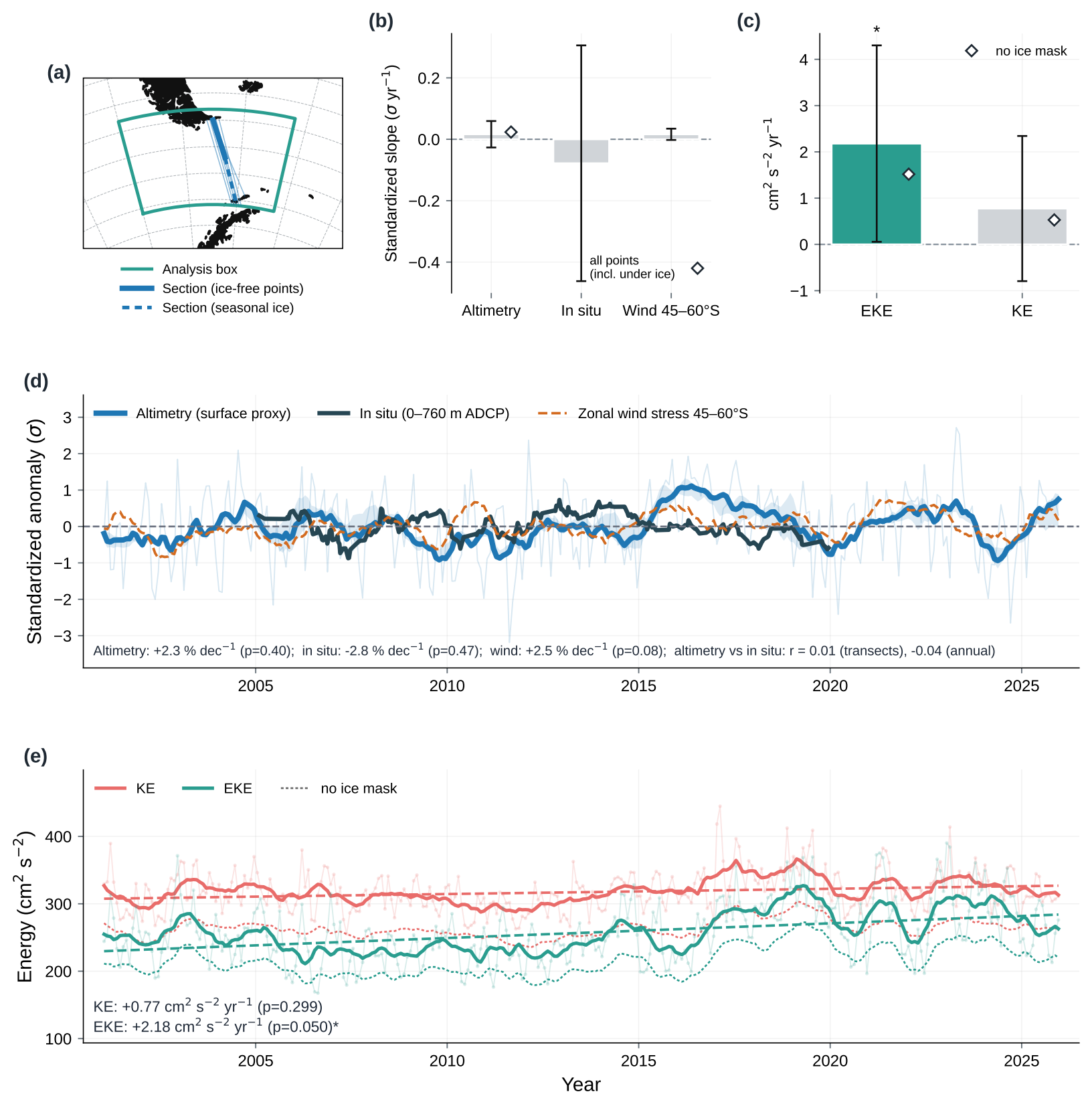

(a)
Analysis box
Section (ice-free points)
Section (seasonal ice)
(b)
Standardized slope (σ yr−1)
0.2
0.0
−0.2
−0.4
all points
(incl. under ice)
Altimetry
In situ
Wind 45–60°S
(c)
*
no ice mask
cm2 s−2 yr−1
4
3
2
1
0
−1
EKE
KE
(d)
Altimetry (surface proxy)
In situ (0–760 m ADCP)
Zonal wind stress 45–60°S
Standardized anomaly (σ)
3
2
1
0
−1
−2
−3
Altimetry: +2.3 % dec−1 (p=0.40); in situ: -2.8 % dec−1 (p=0.47); wind: +2.5 % dec−1 (p=0.08); altimetry vs in situ: r = 0.01 (transects), -0.04 (annual)
2005
2010
2015
2020
2025
(e)
KE
EKE
no ice mask
Energy (cm2 s−2)
400
300
200
100
KE: +0.77 cm2 s−2 yr−1 (p=0.299)
EKE: +2.18 cm2 s−2 yr−1 (p=0.050)*
2005
2010
2015
2020
2025
Year